



Redefining the Boundary Spanning-Environment Relationship

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A review of previous studies on the relationship between boundary spanning activity and environment indicates that the full effects of environment have not been investigated due to the restricted representation of environmental effects. In addition, we argue that the boundary spanning-environment relationship may be moderated by function, hierarchical level, perceived influence, size, and industry. Accordingly, an enlarged model of the boundary spanning-environment relationship is tested in both the high-technology electronics industry and the wood products industry. Results indicate that boundary spanning activity is related to environment, but this relationship appears to vary along dimensions of environment as well as by industry. In addition, size, perceived influence, and function are found to moderate the boundary spanning-environment relationship. Implications for further studies include a reconceptualization of the boundary spanning-environment relationship based on the findings.

Open systems theory suggests that boundary spanning should increase as the environment becomes more uncertain and turbulent (Kast & Rosenzweig, 1979; Katz & Kahn, 1966). An unstable and unpredictable environment poses potential opportunities and threats to the organization's future prosperity and survival, forces which must be dealt with in particular by managers at middle and upper levels in the organization. The essential dilemma faced by managers is the maintenance of a stable and efficient set of internal operations while maintaining a flow of new ideas and innovations for continued improvements. To accomplish this, a rational response by the organization is to increase its efforts to scan, interpret, monitor, and control its environment through formal or informal boundary spanning activities (Thompson, 1967).

In spite of the intuitive appeal of this hypothesis, it has not been conclusively supported in recent empirical research. Leifer and Huber (1977) suggest that or-

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ganizational structure, and not environmental uncertainty, is the primary determinant of boundary spanning. Cox, Hitt, and Stanton (1978) found a relationship between boundary relevance and perceived environmental uncertainty, but found an even stronger relationship between boundary relevance and hierarchical level. Related studies (e.g., Jemison, 1979) indicate that influence on organizational decisions is the key determinant of boundary spanning. These findings can be broadly interpreted as supportive of the view that variations in boundary spanning are best explained by structural or contextual factors (e.g., structure, level, influence) rather than the traditional view that boundary spanning is principally stimulated by environmental uncertainty and change.

This paper reports on an empirical study which tested whether boundary spanning is primarily related to environmental changes or to structural variables. The relationship between boundary spanning and environment is reconceptualized using an enlarged set of environmental measures and provisions for examining the possible confounding effects of structural variables. Samples were taken from two separate industries—wood products and high technology (electronics). Finally, based on our results, a modified view of the boundary spanning-environment relationship is presented.

Reconceptualizing the Boundary Spanning-Environment Relationship

Osborn, Hunt, and Jauch (1980) have suggested that an organization's environment can be represented in two segments. The first is called the general environment and deals with the overall cultural, economic, educational, and legal-political conditions facing particular organizations. The second refers to the specific environment that includes organizations with which the particular organization interacts in its attempt to survive. Munificent or scarce environments arising from changes in the cultural, economic, and other conditions of the general environment create both opportunities and constraints for the focal organization. However, these effects are more directly represented in focal exchanges between the organization and its specific (or task) environment, that is, sectors that affect the goal-setting activities of the focal organization. Exchanges between the focal organization and its specific environment are necessary to overcome inabilities to predict or control correctly the actions of specific sectors of the task environment (Thompson, 1967). These changes are not managed by the organization per se. Rather, the organization relies upon individuals to perform these boundary spanning activities and functions on its behalf (Miles, 1980). The need for these environmental exchanges and transactions directly affects the development of boundary spanning activities by individuals within the focal organization.

Open systems theory implies that boundary spanners need to correctly scan and monitor the activities of the task environment. Failure to scan the environment or a serious misdiagnosis of environmental events could lead to organizational crisis and failure (Starbuck, Greven, & Hedberg, 1978). Even at times when environmental conditions are certain and stable, boundary spanners develop and maintain the environmental niche in a manner conducive to effective

performance. The initial selection of suppliers, distributors, and the like constitutes only the first step in developing a niche. Even with the environment growing at a stable and predictable rate, the degree of interdependence between different sectors of the environment and focal organization may be undergoing fundamental changes. Effective interaction between boundary spanners and the task environment is necessary to develop informed decisions on such changes.

The foregoing discussion suggests three contingencies imposed by the environment on organizations in general, and on boundary spanners in particular: (a) the uncertainty or the unpredictability of task-environmental sectors; (b) the dependence imposed by task-environmental sectors resulting from the inability of organizations to control critical resources; and (c) the relative importance of task environmental sectors in accommodating short- and long-term requirements for the organization to remain effective in its environment. While each contingency is conceptually distinct, their effects on the organization (and boundary spanners) are interactive. For example, the effects of a highly unpredictable task-environmental sector (e.g., labor) that is not considered to be highly important might not be particularly adverse to the organization. Thus, boundary spanners would not focus much effort or activity on this environmental sector.

Recent research suggests, however, that boundary spanning might not result primarily from environmental change but from contextual variables such as size, industry, functional position, level, and influence. Large organizations deal with the external environment differently than small firms, as demonstrated in adjustments made by large and small firms in response to government regulations (Cole & Tegeler, 1978). Hrebiniak and Snow (1980) have also found that managerial perceptions of environmental uncertainty differ across industries, suggesting that industry differences must be specifically considered when interpreting boundary spanning activities or environmental perceptions.

The argument that boundary spanning has some important structural correlates (functional position, hierarchical level, and influence on organizational decisions) is theoretically supported in the other works. Dearborn and Simon (1958) have shown that environmental scanning is related to functional position. That is, certain positions inherently require dealing with the external environment (e.g., salesman, purchasing agent). Similarly, it is generally hypothesized that higher hierarchical positions deal more extensively with the external environment of the organization in comparison to lower level positions that are more oriented to internal operations (Brown, 1966; Parsons, 1960; Thompson, 1967). One can also argue that boundary scanning in certain cases is related to influence. That is, if interaction with an uncertain environment becomes critical for organizational survival, the organization will be very dependent on those involved in that interaction (the boundary spanners) for information, prediction of trends, and control of potential problems in the environment (Crozier, 1964; Hickson, Hinnings, Lee, Schneck, & Pennings, 1971; Pfeffer & Salancik, 1978).

Overall, these arguments provide two possible explanations for the lack of conclusive support for the traditional view that boundary spanning arises principally from environmental changes. First, the full effects of the environment on boundary spanners have not been adequately represented and tested; and second, the

linkage between environment and boundary spanning exists, but is confounded by other variables such as structure, function, size, level, and influence. Accordingly, this study addresses two questions:

1. Is there a relationship between boundary spanning activity and environment (using an expanded set of environmental variables)?
2. How is the relationship between boundary spanning activity and environment affected by contextual variables such as size, function, influence, and hierarchical level?

Method

Sample

Hrebiniak and Snow (1980) found that perceptions of the environment differ across industries, suggesting that industry differences must be considered in boundary spanning studies. Specifically, it is important that the industries exhibit sufficient variation in the study variables. Methodologically, this also indicates that special care be taken to avoid indiscriminate pooling of firms from different industries such that results confuse the underlying relationships.

The industries selected for this study were high-technology electronics and wood products. Available information about these industries at the time the data were collected (winter of 1980) indicated the types of variation we wanted for purposes of our study. The high-technology electronics industry was young, rapidly growing, innovative, facing a highly competitive market; in contrast, the wood products industry was mature, more technologically stable, and coping with recessionary pressures. In particular, wood products firms were faced with problems related to the uncertain supply of raw materials. They were also faced with problems dealing with government regulations. These can be contrasted with the problems faced by the high-technology electronics industry, which centered on capital availability, cost efficiency, and managing of innovation. These differences bear directly on this study of boundary spanning for two reasons. First, both industries were faced with pressures from their task environments, a condition necessary to test conceptual validity of boundary spanning as a research issue. Second, there are notable contrasts in these pressures arising from the industry stage of growth. The need to effectively innovate, for example, is one that would apply more readily to a growing industry such as high-technology electronics in contrast to a mature industry such as wood products. Taken together, these features provide the necessary latitude for examining environmentally related differences in boundary spanning activity.

To ensure the proper classification of firms within industries, standard industrial classification codes (SIC's) were used to separate wood product firms (SIC codes 24 and 26) from high-technology electronic firms (SIC codes 36 and 38). Specific firms from each industry were selected from standard corporation directories (*Standard and Poor's*; *Dun and Bradstreet's Million Dollar Directory*; *American Electronics Association Directory*).

Data Collection

Questionnaires were sent to the chief executive officer or his designated representative for distribution to employees believed to have primary responsibility for dealing with the external environment. Three hundred seventy-nine individuals from 36 companies participated in the study, for a 57% response rate (287 individuals from 24 high-technology electronics companies, for a 54% response rate; 92 individuals from 12 wood products firms, for a 68% response rate). Responses were also analyzed to check for sample size bias for respondents and nonrespondents, and tests were insignificant ($X^2 = .7557$, $df = 4$, $p = ns$).

Research Instruments

Boundary spanning was measured using Miles' (1980) set of eight boundary spanning activities, slightly modified by splitting his "transacting" activity into two parts: acquiring and distributing. Since open systems theory distinguishes the input and output activities of the organization (see Kast & Rosenzweig, 1979), it seemed these two activities would capture more precisely potential differences in boundary spanning. This orientation is also consistent with the work of Stewart (1982), who describes the demands (e.g., minimum criteria, work diversity), constraints (e.g., resource limitations), choices (e.g., techniques to accomplish the job), and role assignments of managers.

The boundary spanning activities were factor analyzed (using varimax rotation) and all eight boundary spanning activities loaded significantly on one factor with loadings of .70, .57, .76, .75, .70, .75, .51, and .53. Therefore, an overall boundary spanning score (coefficient alpha = .87) was calculated from these eight activities and used in subsequent tests.

Environmental effects were measured by asking each respondent to answer, on a 5-point scale, the following questions as they applied to each of thirteen sectors of the task environment: "How important is this sector in your job?", "To what extent can you accurately predict the actions of this sector on your organization?", and "To what extent can you control the actions of this sector in your organization?" The thirteen sectors were adapted from Duncan (1972) and included product/service distributors, product/service users, suppliers, competitors, government regulatory agencies, unions, and the like. When factor analysis (with varimax rotation) was conducted, three factors emerged and were interpreted in terms of importance, predictability, and control. Thus, assignment variables obtained by averaging raw scores for each dimension were used in subsequent analysis. Coefficient alphas for each of these dimensions were .79 for importance, .86 for predictability, and .83 for control.

Influence on organizational decisions was measured using thirteen potential types of decisions (adapted from Bacharach & Aiken, 1976; Jemison, 1979). Respondents first indicated type of formal authority for each decision on a 5-point scale; then they indicated how often they could change, modify, or influence the same decisions, regardless of their formal authority for them, on a similar 5-point scale. Results indicated that the two measures were highly intercorrelated ($r =$

.84, $p < .001$), and an overall influence scale was then calculated, with coefficient alpha of .92.

Finally, as boundary spanning, perceived environmental uncertainty, and perceived influence were obtained from a single source, special measures were made to prevent artifactual relationships arising principally from response bias. In addition to multi-item scales, the questions were formatted in matrix scores and, in at least one instrument, reverse scoring was employed. Two additional cross-checks were used: (a) perceived influence (obtained from self-reports) was correlated with hierarchical level (obtained from organizational charts and position titles), and this relationship was significant ($F = 34.78$, $p < .001$); (b) the overall measure of boundary spanning was also correlated with the percentage of time reportedly spent in scanning activities (a measure that was included in the study but not part of this research), yielding a significant, positive relationship ($r = .51$, $p < .001$).

The variables of functional category and hierarchical level were measured by asking the respondents for their job titles and those of their immediate supervisors and direct subordinates. Seven functional categories—administrative, engineering, financial, manufacturing, marketing, personnel, and purchasing—and a three-level hierarchical classification (see Kast & Rosenzweig, 1979; Parsons, 1960) were used. Validity of the assigned hierarchical levels was further assessed by asking respondents to place themselves in one of the three hierarchical levels and to indicate the number of administrative levels separating their positions from the top level of the organization. The results revealed a very strong agreement between the assigned and self-reported indicators of hierarchical level ($X^2_4 = 107.53$, $p < .0001$; $X^2_{14} = 130.16$, $p < .001$, respectively).

Finally, organization size and industry were established for each respondent by pre-coding the questionnaires sent to each firm so responses could be identified by the company from which they came. Company size (e.g., number of employees) was obtained from standard corporation directories (e.g., *Standard and Poor's*). As several of the variables (e.g., functional position, hierarchical level) were nominally or ordinally scaled, dummy indicators were created to represent hierarchical levels and functional positions. Since several investigators have found curvilinear relationships between size and other organizational characteristics (see Pugh, Hickson, Hinnings, & Turner, 1969), the logarithm of size was used as the size control variable in this study. Consistent with the findings of Hrebiniak and Snow (1980), data from each industry were analyzed separately.

Results and Discussion

Results are presented in three sections. First, we present an intercorrelation matrix to provide an overall perspective of the relationship between our study variables. Second, we analyze how boundary spanning is related to environmental variables, with provisions to control for the possible confounding effects of contextual variables. The third section discusses the findings in relation to the research hypothesis.

Overall Variable Tendencies

The means, standard deviations, and zero order correlations for the study variables in the two industries are presented in Tables 1 and 2.

Results indicate that boundary spanning activity is positively related to environmental variables in both industry samples at a significant level ($p = .001$). However, the data also show that boundary spanning is significantly related to structural or contextual variables: specifically, to influence ($r = .223$) and to function ($r = .358$) in the electronics sample and to influence ($r = .281$) and to size ($r = -.277$) in the wood products sample. Furthermore, there are several significant correlations within the environmental and contextual variables, as well as across these sets of variables. In both industry samples, perceptions of environmental importance were positively related to environmental predictability (electronics, $r = .498$, $p < .001$; wood products, $r = .739$, $p < .001$). Positive relationships were also found in both samples between environmental importance and control (electronics, $r = .419$, $p < .001$; wood products, $r = .664$, $p < .001$) and between perceptions of environmental predictability and control (electronics, $r = .480$, $p < .001$; wood products, $r = .666$, $p < .001$). These findings indicate that environmental importance, predictability, and control are very much interrelated, which suggests some underlying relationships. Therefore, subsequent investigations of any one of these variables should control carefully

Table 1
Means, Standard Deviations, and Coefficient Alphas
for the Study Variables

Major Variables	Electronics		Wood Products		Coefficient
	(usable $n = 278$)		(usable $n = 89$)		Alpha
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	r_{sb}
Overall Boundary Spanning Activity	3.253	.8528	2.982	.7936	.87
Perceived Task Environment Importance	3.082	.6492	3.119	.7452	.79
Predictability	2.741	.6482	2.710	.8368	.86
Control	2.162	.6174	2.266	.7078	.83
Control/Contextual Variables					
Perceived Influence on Decisions	2.802	.7097	2.793	.7464	.92
Organization Size (raw)	1173	2134	1789	1065	--
Hierarchical Level ^a	<i>Cases</i>	<i>% of Total</i>	<i>Cases</i>	<i>% of Total</i>	
Top	38	13.7	14	15.7	
Middle	114	41.0	50	56.2	
Bottom	126	45.3	25	28.1	
Functional Area ^a					
Administrative	19	6.9	7	7.9	
Engineering	53	19.1	7	7.9	
Financial	32	11.5	9	10.1	
Manufacturing	47	16.9	28	31.4	
Marketing	96	34.5	9	10.1	
Personnel	17	6.1	5	5.6	
Purchasing	14	5.0	24	27.0	

^aThe distribution for categorical variables is expressed in terms of percentage to total.

Table 2
Zero-Order Correlations Among Study Variables^a

Study Variables	1	2	3	4	5	6	7 ^b	8 ^b
1. Boundary Spanning Activity <i>Environmental Dimensions</i>	--	.378***	.420***	.223*	.311**	-.356***	.313*	.322
2. Perceived Importance	.352***	--	.739***	.664***	.391***	-.248*	.256	.561***
3. Perceived Predictability	.226***	.498***	--	.666***	.281**	-.277**	.161	.434**
4. Perceived Control	.282***	.419***	.480***	--	.371***	-.096	.205	.504***
<i>Contextual Variables</i>								
5. Perceived Influence	.223***	.281***	.259***	.549***	--	-.075	.543***	.474**
6. Organization Size (log 10)	.073	-.072	-.228***	-.188**	-.123*	--	.206	.372
7. Hierarchical Level ^b	.129	.164*	.237***	.410***	.567***	.278***	--	.***
8. Functional Area ^b	.358***	.350***	.294***	.366***	.472***	.248**	.***	--

^aThe upper right-hand matrix consists of zero-order correlations for the wood products sample ($n = 92$); the lower left-hand matrix consists of zero-order correlations for the high-technology electronics sample ($n = 278$).

^bThe values reported for hierarchical level and functional area are multiple R values derived when simple regressions (one dependent variable and one independent variable) were calculated. Since hierarchical level and functional area were both treated as nominal variables, no signs can be attached to these values, yet they can be tested for significance.

^cSince hierarchical level and functional area were both nominally scaled variables, it was impossible to calculate either a correlation coefficient or a multiple R value to represent the relationship between them. Chi-square tests revealed that the two variables were significantly interrelated in both samples (wood products, $X^2_{12} = 28.681$, $p = .005$; electronics, $X^2_{12} = 107.666$, $p = .001$).

* $p = .05$. ** $p = .01$. *** $p = .001$.

for effects caused by the other two environmental dimensions. These interrelationships make it difficult to interpret underlying effects of these variables on boundary spanning and emphasize the need to examine hypothesized relationships using partial-order effects (Cohen & Cohen, 1975).

Boundary Spanning, Environmental Effects, and Contextual Variables

Regression analysis was used to test for the relationships between boundary spanning activity and dimensions of the environment, while simultaneously controlling for the possible confounding effects of contextual variables. The separate or singular effects of the contextual variables were then determined by developing full and partial regression models and comparing the sum of squares error associated with the difference between these models (Draper & Smith, 1966).

Results of this analysis indicate that boundary spanning activity is significantly related to environmental effects even after controlling for the effects of the contextual variables, although the relationship is different for each industry. In the high-technology electronics industry, boundary spanning is significantly related to perceived environmental importance ($F = 19.709, p < .001$) and perceived environmental control ($F = 3.909, p < .05$), but not to perceived predictability. In the wood products industry, boundary spanning is significantly related to perceived environmental predictability ($F = 3.032, p < .09$) but not to perceived importance and perceived control.

In examining the relationship between boundary spanning activity and the contextual variables, some relationships appear to be consistent for the two industries. Specifically, boundary spanning is significantly related to perceived influence (electronics, $F = 6.603, p < .01$; wood products, $F = 4.496, p < .05$), and functional area (electronics, $F = 10.492, p < .001$; wood products, $F = 2.253, p < .05$). Boundary spanning activity is positively related to size in the high-technology electronics sample ($F = 3.148, p < .08$), but negatively related to size in the wood products sample ($F = 8.304, p < .006$). In both industry samples, boundary spanning activity is not significantly related to hierarchical level.

In addition, one can argue that the interaction effects between the environmental variables and with the contextual variables can be meaningful in explaining boundary spanning activity. Duncan (1973) has argued that decision-making structures can be better explained by examining the uncertainty faced by the decision-making units in conjunction with the amount of control these units have over their environment. In this study, these can be represented by considering the joint product term produced by multiplying predictability by control. Moderated regression was used to evaluate the significance of the joint term on the criterion variable (boundary spanning).

Analysis showed no significant interactive effects for the high-technology electronics sample, while the significant interactions for the wood products sample were limited to environmental dimensions, that is, the joint effects of importance and predictability ($F = 4.566, p < .05$), and predictability and control ($F = 3.873, p < .06$).

Discussion

Several important conclusions may be drawn from the results obtained. Perhaps most basically, a clear relationship does indeed exist between boundary spanning activity and environment, but this relationship appears to vary with environmental dimensions as well as with industry. The relationship between boundary spanning activity and environment was better explained by the use of importance, predictability, and control as composite measures of environment. In the wood products sample, boundary spanning activity increases with the ability to predict the actions of task environmental sectors, although the relationship is relatively weak. The differential impact of environmental dimensions (i.e., importance, predictability, and control) might help clarify some of the conflicting findings in previous research. Since Leifer and Huber (1977) used only a predictability dimension in their environmental uncertainty measure, they might have understated the relationship between boundary spanning and the environment. Our results suggest that future studies of the relationship between perceived environment and boundary spanning may be spurious unless the importance and control of environmental sectors are considered along with environmental predictability (uncertainty).

Some findings are more interpretable when industry differences are taken into account. Among wood products firms, it is possible that boundary spanning activity may be routinized as part of organizational structure. As an industry moves from growth to mature stages of growth, it is reasonable to expect more stability and greater learning among industry participants (Porter, 1980). As these firms learn to anticipate environmental changes over time, boundary spanning may be institutionalized in terms of standard operating procedures and organizational structure. Consistent with arguments made by Galbraith (1977) and Jemison (1979), it is possible that boundary spanners in these firms view themselves as becoming more influential in the organization.

Boundary spanning activity in the high-technology electronics sample significantly increases with the importance and control ascribed to certain task-environmental sectors, but is not related to predictability. Porter (1980) has suggested that growth stages of an industry are often accompanied by widening buyer groups with a greater emphasis on product innovation, quality, and reliability. In effect, buyers' and suppliers' actions may be difficult to predict (e.g., there are multiple movements in and out of the industry), and firms would fare better if they concentrated on controlling sectors of their task environment (e.g., special customers, buyers, etc.) that are considered to be important to the organization.

A perplexing finding is the lack of statistical relationship between boundary spanning activity and hierarchical level. This is surprising in view of the attractiveness of the conceptual relationship: that scanning and monitoring of the task environment should reside at the institutional level. There are two possible explanations for this. First, in highly decentralized organizations, it is possible that transactions with the task environment are prevalent throughout the organization. Therefore, boundary spanning activity would extend across hierarchical levels. A second possible reason is the effect that social desirability might have on this research. Respondents might wish (consciously or unconsciously) to present a pos-

itive view of their function by overstating their information-gathering activities. Since we do not have more objective data with which to corroborate reported behaviors, we cannot resolve this issue at this time. Parenthetically, our review of previous studies that claimed a relationship between level and boundary spanning did not shed much insight into this issue. Both Cox et al. (1978) and Miles (1976) used "boundary relevance," not boundary spanning, as the dependent variable, and Leifer and Huber (1977) cautioned that their data on hierarchical level might have been too restrictive. Clearly, this is an area which requires more empirical attention in the future.

In addition to the implications for research, these findings also indicate the importance of boundary spanning activities in the practical sense. They give us a clearer view of the complexity involved in this basic managerial activity and infer some directions concerning the particular roles, information needs, and types of boundary spanning activities that managers should encourage.

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